1. Construct and communicate explanations and arguments in the context of the problem (Point 7 in the list)

This is the core requirement of this question. Students need to make a clear judgment (Yes or No) about the tester's claim and provide a mathematical argument supported by calculations for their answer.

- **Specific Analysis:** Students must calculate the average number of faulty video players and audio players per day.
  - Video faults: $2000 \times 5\% = 100$
  - Audio faults: $6000 \times 3\% = 180$
  - Students need to compare these two numbers (100 and 180), realize that the number of video players sent for repair is less than that of audio players, and thus the tester's claim is **incorrect**.
  - They need to clearly show these calculation steps and use text to explain how the comparison result refutes the tester's claim, thereby constructing a complete mathematical argument.

2. Identify the mathematical aspects of a real-world problem situation including identifying the significant variables (Point 6 in the list)

Before starting the calculations, students must extract relevant mathematical information from the real-world context.

- **Specific Analysis:** Students need to identify key variables from the table: the number of video players (2000), the number of audio players (6000), the video fault rate (5%), and the audio fault rate (3%). They must ignore irrelevant information and understand the meaning of terms like "average" and "fault rate" within the problem context.

3. Interpret a mathematical result back into the real world context (Point 1 in the list)

The final results must be interpreted back into the actual scenario.

- **Specific Analysis:** After obtaining the numbers 100 and 180, students need to interpret them as the "number of video players sent for repair per day" and the "number of audio players sent for repair per day," and use this to judge whether the tester's claim of "more video units" is reasonable.

1. 构建并传达基于问题背景的解释和论证 (列表中的第 7 点)

这是此题的核心要求。学生需要对测试员的说法做出明确判断（是或否），并通过计算提供数学论证来支持他们的答案 (p. 1)。

- **具体分析：** 学生必须计算出每天视频播放器和音频播放器的平均故障数量。
  - 视频故障数量：$2000 \times 5\% = 100$ (p. 1)
  - 音频故障数量：$6000 \times 3\% = 180$ (p. 1)
  - 学生需要比较这两个数字（100 和 180），发现视频播放器送修的数量少于音频播放器，因此测试员的说法是**不正确**的 (p. 1)。
  - 他们需要清晰地展示这些计算步骤，并用文字说明比较结果如何反驳测试员的说法，从而构建一个完整的数学论证。

2. 识别现实世界问题情境的数学方面，包括识别重要变量 (列表中的第 6 点)

在开始计算之前，学生必须从现实世界的情境中提取相关的数学信息。

- **具体分析：** 学生需要从表格中识别关键变量：视频播放器的数量（2000）、音频播放器的数量（6000）、视频故障率（5%）和音频故障率（3%）(p. 1)。忽略不相关的信息，并理解“平均”、“故障率”等术语在问题背景下的含义。

3. 将数学结果解释回现实世界的情境中 (列表中的第 1 点)

最终的结果必须回到实际场景中进行解读。

- **具体分析：** 得出 100 和 180 这两个数字后，学生需要将它们解释为“每天送修的视频播放器数量”和“每天送修的音频播放器数量”，并据此判断测试员的“视频数量更多”的说法是否合理 (p. 1)。